

Fall 2025.

Physics 509. Quantum Theory I.
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Homework # 1.

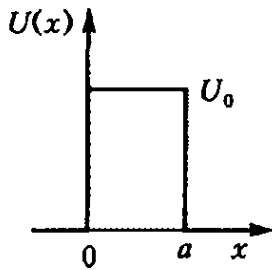
Due to September 24th, 2025.

You can use any source to solve the problems. However, your solutions should be presented in a very detailed and logical way. You need to understand every step of your solution and be able to explain it. Provide me with your code and final graphs when you use computational methods.

1. Find the energy levels and normalized wave functions for a particle in an infinite potential well

$$U(x) = \begin{cases} 0, & 0 < x < a \\ \infty, & x < 0, x > a \end{cases}$$

2. Calculate a transmission coefficient for a rectangular barrier if $E > U_0$. See the figure below.



3. Find the energy levels and normalized wave functions for a particle in an infinite potential well

$$U(x) = \begin{cases} 0, & -a < x < a \\ \infty, & x < -a, x > a \end{cases}$$

See **Desai Section 8.1.1**. Pay attention to existence of two types of solutions, such as odd and even.

4. Now find the energy levels of a finite potential well

$$U(x) = \begin{cases} 0, & -a < x < a \\ U_0, & x < -a, x > a \end{cases}$$

. See **Desai Section 8.1.2**. Use any numerical method to solve corresponding transcendental equations for the energy levels. Again, pay attention to existence of two types of solutions, such as odd and even.

5. Consider the potential from **Desai Section 8.4**. Go through all calculations in detail and find the energy levels numerically.

6. Find energy levels and wave functions of the double-well δ -potential

$$U(x) = -\frac{\hbar^2}{m}\alpha(\delta(x-a) + \delta(x+a))$$

Hint: The potential is an even function. Therefore, you should consider two cases of even and odd wave functions separately.

7. Consider a full version of Kronig-Penney model with a periodic rectangular potential. See, for example, **Desai Section 8.5**. Find the equation for a dispersion law and use your computational skills to demonstrate the band structure presence. You can use Mathematica for long calculations.